

A/B Interface Chain Model

Theoretical Background, Hamiltonian Definition, Time Propagation, and User Guide

AB Interface Toy-Model Software Documentation

Version for the no-impact GUI package

Contents

1 Purpose of the Software

This software implements a compact density-matrix toy model for short-time laser-driven carrier and spin redistribution at a non-matching interface between two materials, denoted by A and B . The model is designed for qualitative mechanism analysis rather than quantitative materials prediction. It is useful when one wants to test whether a proposed sequence of optical excitation, spin-orbit-induced spin mixing, interlayer transfer, and intramaterial relaxation can generate a target population-flow pattern.

The current main mechanism is the chain

$$A_v \uparrow \xrightarrow{\text{laser}} A_c \uparrow \xrightarrow{\text{SOC mixing in } A_c} A_c \downarrow \xrightarrow{\text{spin-conserving interlayer transfer}} B_c \downarrow \xrightarrow{\text{intramaterial relaxation in } B} B_v \downarrow. \quad (1)$$

This mechanism replaces the earlier, less transparent direct interface-SOC transition

$$A_c \uparrow \leftrightarrow B_v \downarrow, \quad (2)$$

which bundled material change, spin flip, and band relaxation into a single phenomenological matrix element. In the present implementation, spin flip is produced primarily by material-internal SOC mixing, while the interlayer transfer is spin-conserving.

The code generates time-dependent projected occupations, magnetic moment proxies, static-eigenbasis occupations, state-resolved occupation-change diagrams, CSV data, and a graphical user interface. It is packaged as a Python project and can be run either from the command line or through a Tkinter desktop GUI.

2 Physical Model Overview

The model separates the dynamics into coherent Hamiltonian evolution and incoherent post-pulse rate processes. The time-dependent density matrix is denoted by $\rho(t)$. The coherent part is governed by

$$H(t) = H_{\text{static}} + H_{\text{laser}}(t), \quad (3)$$

where

$$H_{\text{static}} = H_{\text{bands}} + H_{\text{SOC,mix}} + H_{\text{interlayer}}. \quad (4)$$

The model contains the following physical components:

1. **No-SOC non-matching bands:** materials A and B have independent gaps, energy offsets, and band widths.
2. **SOC-derived diagonal splitting:** spin splitting is not an independent input; it is computed from material-internal SOC strengths $\lambda_{\text{soc},A}$ and $\lambda_{\text{soc},B}$.
3. **Material-internal SOC mixing:** off-diagonal couplings mix $\uparrow$ and $\downarrow$ components within the same material and band manifold.
4. **Spin-conserving interlayer hopping:** the primary interlayer process is $A_c s \leftrightarrow B_c s$ with $s = \uparrow, \downarrow$.
5. **Laser excitation:** the optical pulse drives spin-conserving intramaterial $v \leftrightarrow c$ transitions.
6. **Post-pulse rate processes:** after the laser pulse, the model applies spin-conserving interlayer downhill transfer and intramaterial $c \rightarrow v$ relaxation.

The former impact-excitation branch has been removed from the main model. Without explicit electron-electron Coulomb matrix elements and a controlled bath spectral density, impact excitation is too phenomenological for the present short-time chain mechanism and obscures pathway interpretation.

3 Single-Particle Basis

Each material and spin block contains N single-particle states. The total Hilbert-space dimension is

$$D = 4N. \quad (5)$$

The basis ordering is

$$A_{\uparrow}, \quad A_{\downarrow}, \quad B_{\uparrow}, \quad B_{\downarrow}. \quad (6)$$

In each block, the first $N/2$ states are valence-like and the last $N/2$ states are conduction-like. Therefore N must be even. In index form,

Block	Index range	Meaning
$A_{\uparrow}$	$0, \dots, N - 1$	material A , spin up
$A_{\downarrow}$	$N, \dots, 2N - 1$	material A , spin down
$B_{\uparrow}$	$2N, \dots, 3N - 1$	material B , spin up
$B_{\downarrow}$	$3N, \dots, 4N - 1$	material B , spin down

Within any block,

$$0, \dots, N/2 - 1 \quad \text{are valence-like states,} \quad (7)$$

while

$$N/2, \dots, N - 1 \quad \text{are conduction-like states.} \quad (8)$$

The software uses helper functions in `basis.py` to avoid hard-coded manual indexing.

4 No-SOC Band Energies

The no-SOC band structure is specified by material-dependent gaps, offsets, and bandwidths. For material $X \in \{A, B\}$, the valence-band maximum and conduction-band minimum before SOC splitting are approximately

$$E_{X,\text{VBM}}^{(0)} = \text{offset}_X - \frac{1}{2}\text{gap}_X, \quad (9)$$

$$E_{X,\text{CBM}}^{(0)} = \text{offset}_X + \frac{1}{2}\text{gap}_X. \quad (10)$$

The valence ladder is constructed downward from the VBM:

$$E_{X,v,m}^{(0)} = E_{X,\text{VBM}}^{(0)} - \epsilon_m^{X,v}, \quad (11)$$

where $\epsilon_m^{X,v}$ spans the valence bandwidth. The conduction ladder is constructed upward from the CBM:

$$E_{X,c,n}^{(0)} = E_{X,\text{CBM}}^{(0)} + \epsilon_n^{X,c}, \quad (12)$$

where $\epsilon_n^{X,c}$ spans the conduction bandwidth.

The default alignment is staggered so that material A has the lower conduction-band edge and material B has the higher valence-band edge. This is suitable for testing an interfacial carrier-transfer scenario in which excited carriers can leave A_c and populate B_c , followed by relaxation into B_v .

5 SOC-Derived Diagonal Splitting

The code does not expose independent spin-splitting parameters. Instead, the diagonal SOC splitting is derived from material SOC strengths:

$$\lambda_{\text{soc},A}, \quad \lambda_{\text{soc},B}. \quad (13)$$

The convention used by the code is

$$E_{\uparrow} = E_0 + \frac{\Delta}{2}, \quad E_{\downarrow} = E_0 - \frac{\Delta}{2}. \quad (14)$$

For the default chain mechanism, spin-down CBM lies above spin-up CBM. Therefore the internal sign convention is

$$\Delta_A = -|\lambda_{\text{soc},A}|, \quad \Delta_B = -|\lambda_{\text{soc},B}|. \quad (15)$$

With this sign choice,

$$E_{X,\downarrow} = E_0 + \frac{|\lambda_{\text{soc},X}|}{2}, \quad E_{X,\uparrow} = E_0 - \frac{|\lambda_{\text{soc},X}|}{2}. \quad (16)$$

This makes the spin-down band edge higher than the spin-up band edge for both A and B , which is the qualitative ordering assumed in the default scenario.

6 Hamiltonian Terms

6.1 Band Hamiltonian

The diagonal band Hamiltonian is

$$H_{\text{bands}} = \sum_i E_i |i\rangle \langle i|, \quad (17)$$

where $|i\rangle$ is a diabatic basis state such as $|A, c, n, \uparrow\rangle$ or $|B, v, m, \downarrow\rangle$. The energies E_i include the no-SOC band positions and the SOC-derived diagonal splitting.

6.2 Material-Internal SOC Mixing

Diagonal SOC splitting alone does not transfer population between spin channels. To realize the step

$$A_c \uparrow \rightarrow A_c \downarrow, \quad (18)$$

the model includes off-diagonal material-internal SOC mixing. In the simplest same-band-index form,

$$H_{\text{SOC,mix}} = \sum_{X=A,B} \sum_{b=v,c} \sum_m \Omega_{X,b} |X, b, m, \downarrow\rangle \langle X, b, m, \uparrow| + \text{h.c.} \quad (19)$$

The corresponding software parameters are

Parameter	Meaning
<code>soc_mix_cb_A</code>	A conduction-band up/down SOC mixing
<code>soc_mix_cb_B</code>	B conduction-band up/down SOC mixing
<code>soc_mix_vb_A</code>	A valence-band up/down SOC mixing
<code>soc_mix_vb_B</code>	B valence-band up/down SOC mixing

The conduction-band mixing parameters are especially important because they allow the chain process

$$A_c \uparrow \rightarrow A_c \downarrow. \quad (20)$$

They also allow SOC-mixed B_c states, so occupation accumulated in $B_c \downarrow$ can indirectly affect $B_c \uparrow$ optical availability through mixed eigenstates.

6.3 Spin-Conserving Interlayer Hopping

The interlayer Hamiltonian is spin-conserving. The main term is conduction-conduction hopping,

$$H_{\text{interlayer}}^{cc} = \sum_{m,n,s} T_{mn}^{cc} |B, c, n, s\rangle \langle A, c, m, s| + \text{h.c.} \quad (21)$$

The matrix element is modeled as

$$T_{mn}^{cc} = t_{AB} F_{\text{band}}(m, n) F_E(E_{A,c,m,s} - E_{B,c,n,s}). \quad (22)$$

Here t_{AB} is the interlayer hopping scale, F_{band} is a band-index overlap filter, and F_E is an energy-mismatch filter. The band-index filter is

$$F_{\text{band}}(m, n) = \exp \left[-\frac{(d_m - d_n)^2}{2w_{\text{inter}}^2} \right], \quad (23)$$

where the distances are measured from the relevant band edge. The energy filter is

$$F_E(\Delta E) = \exp \left[-\frac{(\Delta E)^2}{2\sigma_{\text{hyb}}^2} \right]. \quad (24)$$

The corresponding parameters are `tAB`, `interface_band_width`, and `hybrid_energy_width`. An optional valence-valence hopping term can be enabled by setting `tAB_vv` to a nonzero value:

$$H_{\text{interlayer}}^{vv} = \sum_{m,n,s} T_{mn}^{vv} |B, v, n, s\rangle \langle A, v, m, s| + \text{h.c.} \quad (25)$$

By default `tAB_vv=0`.

7 Laser Pulse and Optical Coupling

The laser drives spin-conserving intramaterial valence-conduction transitions:

$$X_v s \leftrightarrow X_c s, \quad X \in \{A, B\}, \quad s \in \{\uparrow, \downarrow\}. \quad (26)$$

The laser Hamiltonian is

$$H_{\text{laser}}(t) = F(t) \sum_{X,m,n,s} d_{mn}^X |X, c, n, s\rangle \langle X, v, m, s| + \text{h.c.} \quad (27)$$

7.1 Pulse Shape

In full-carrier mode,

$$F(t) = A_0 \sin^2 \left(\frac{\pi t}{T_{\text{pulse}}} \right) \cos \left(\frac{\omega t}{\hbar} \right), \quad 0 < t < T_{\text{pulse}}. \quad (28)$$

Outside the pulse window, $F(t) = 0$. In RWA mode, the fast carrier is removed and the drive uses only the envelope.

7.2 Optical Matrix Elements

The optical matrix element is modeled as

$$d_{mn}^X = d_0^X F_{\text{res}}(E_c - E_v) F_{\text{edge}}(m, n). \quad (29)$$

The resonance factor is

$$F_{\text{res}} = \exp \left[-\frac{(E_c - E_v - \hbar\omega)^2}{2\sigma_{\text{opt},X}^2} \right]. \quad (30)$$

The edge-overlap factor uses distances from CBM and VBM. If $d_c = 0$ denotes the CBM and $d_v = 0$ denotes the VBM, then

$$F_{\text{edge}} = \exp \left[-\frac{(d_c - d_v)^2}{2w_{\text{opt}}^2} \right]. \quad (31)$$

This definition intentionally couples CBM-like states most strongly to VBM-like states and avoids the earlier unphysical pairing of CBM with the deepest valence states.

8 Initial State

The initial state is not a manually filled diabatic valence configuration. Instead, the software constructs the full static Hamiltonian

$$H_{\text{static}} = H_{\text{bands}} + H_{\text{SOC,mix}} + H_{\text{interlayer}}, \quad (32)$$

diagonalizes it,

$$H_{\text{static}} |\phi_n\rangle = \varepsilon_n |\phi_n\rangle, \quad (33)$$

and fills the lowest $2N$ static eigenstates:

$$\rho(0) = \sum_{n=1}^{2N} |\phi_n\rangle \langle \phi_n|. \quad (34)$$

This avoids a spurious quench at $t = 0$ and ensures that the initial density matrix already includes static interlayer and SOC hybridization.

9 Coherent Time Propagation

During each time step, the Hamiltonian is evaluated at the midpoint:

$$H_{\text{mid}} = H \left(t + \frac{\Delta t}{2} \right). \quad (35)$$

The unitary propagator is

$$U(t + \Delta t, t) = \exp \left[-\frac{i}{\hbar} H_{\text{mid}} \Delta t \right]. \quad (36)$$

The density matrix is updated by

$$\rho(t + \Delta t) = U \rho(t) U^\dagger. \quad (37)$$

The implementation computes the exponential by diagonalizing the Hermitian midpoint Hamiltonian:

$$H_{\text{mid}} = V \epsilon V^\dagger, \quad U = V \exp \left[-\frac{i\epsilon \Delta t}{\hbar} \right] V^\dagger. \quad (38)$$

For the small matrices used in this toy model, direct dense diagonalization is transparent and sufficiently fast.

10 Post-Pulse Incoherent Rate Processes

After the pulse, two phenomenological rate mechanisms are applied.

10.1 Spin-Conserving Interlayer Downhill Transfer

The software constructs interlayer transfer channels from the spin-conserving interlayer Hamiltonian. A channel $i \rightarrow j$ is allowed if

$$E_i > E_j \quad (39)$$

and

$$\langle j | H_{\text{interlayer}} | i \rangle \neq 0. \quad (40)$$

The channel rate is

$$\Gamma_{i \rightarrow j}^{\text{inter}} = W_{\text{downhill}} \frac{|\langle j | H_{\text{interlayer}} | i \rangle|^2}{t_{AB}^2}. \quad (41)$$

This describes bath-assisted downhill transfer through spin-conserving interlayer hopping. It is not an interface SOC spin-flip term.

10.2 Intramaterial $c \rightarrow v$ Relaxation

Intramaterial relaxation channels have the form

$$X_c s \rightarrow X_v s. \quad (42)$$

The channel rate is

$$\Gamma_{c \rightarrow v}^X = W_{\text{intra}, X} M_{vc}^2 J(E_c - E_v). \quad (43)$$

Here M_{vc} reuses the optical matrix-element hierarchy, so different conduction-valence pairs relax at different rates. The bath-like spectral weight is

$$J(\Delta E) = \frac{\Delta E}{E_{\text{scale}}} \exp\left(-\frac{\Delta E}{E_{\text{scale}}}\right), \quad (44)$$

where E_{scale} is estimated from the material gap and bandwidths.

10.3 Rate Update Rule

All one-electron rate channels use an exponential probability

$$p = 1 - \exp(-\Gamma \Delta t). \quad (45)$$

The transferred occupation is

$$\Delta n = p n_i (1 - n_j). \quad (46)$$

The factor $n_i(1 - n_j)$ implements Pauli blocking. This update keeps occupations bounded between 0 and 1 and eliminates the need for a user-facing collision-fraction stability parameter.

11 Observables and Output Files

The software outputs several time-dependent observables. The most important are:

- material- and spin-projected occupations: $N_{A\uparrow}$, $N_{A\downarrow}$, $N_{B\uparrow}$, $N_{B\downarrow}$;
- pathway-resolved occupations: $A_v \uparrow$, $A_c \uparrow$, $A_c \downarrow$, $B_c \downarrow$, $B_v \downarrow$, and $B_c \uparrow$;
- total spin projections $N_\uparrow$ and $N_\downarrow$;

- a magnetic-moment proxy $M = N_{\uparrow} - N_{\downarrow}$;
- valence and conduction occupation sums;
- static eigenbasis low/high occupations;
- state-resolved occupation changes relative to pulse end.

For output prefix `interface_chain_model`, the standard files are

```
interface_chain_model_projected_material_spin.png
interface_chain_model_pathway_projected_occupations.png
interface_chain_model_projected_spin.png
interface_chain_model_projected_magnetic_moment.png
interface_chain_model_projection_vs_eigen_occupation.png
interface_chain_model_relaxation_diagnostics.png
interface_chain_model_observables.csv
interface_chain_model_state_occupation_change_from_pulse_end_levels.png
interface_chain_model_state_occupation_change_from_pulse_end_levels.csv
```

The pathway-resolved plot is the most direct diagnostic for the proposed chain mechanism.

12 Parameter Reference

12.1 Band Parameters

Parameter	Meaning
N	Number of states per material and spin block. Must be even.
gap_A, gap_B	No-SOC band gaps of materials <i>A</i> and <i>B</i> .
offset_A, offset_B	Rigid energy shifts of the no-SOC spectra.
bandwidth_v_A, bandwidth_v_B	Valence bandwidths.
bandwidth_c_A, bandwidth_c_B	Conduction bandwidths.

12.2 SOC Parameters

Parameter	Meaning
lambda_soc_A, lambda_soc_B	Material-internal SOC scales used to derive diagonal spin splitting.
soc_mix_cb_A, soc_mix_cb_B	Off-diagonal up/down SOC mixing in conduction bands.
soc_mix_vb_A, soc_mix_vb_B	Off-diagonal up/down SOC mixing in valence bands.

12.3 Interlayer Parameters

Parameter	Meaning
tAB	Spin-conserving $A_c s \leftrightarrow B_c s$ hopping scale.

tAB_vv	Optional spin-conserving $A_v s \leftrightarrow B_v s$ hopping scale.
interface_band_width	Band-index overlap width for interlayer hopping.
hybrid_energy_width	Energy mismatch width for interlayer hopping.

12.4 Laser Parameters

Parameter	Meaning
A0	Laser amplitude in model units.
omega_eV	Photon energy $\hbar\omega$ in eV.
pulse_duration	Pulse duration in fs.
carrier	Either full or rwa .
dA0, dB0	Optical dipole scales for materials A and B .
optical_energy_width	Fallback optical resonance width.
optical_energy_width_A, optical_energy_width_B	Material-specific optical widths. Blank or None means use the fallback.
band_overlap_width	Band-edge overlap width for optical and relaxation matrix elements.

12.5 Rate and Time Parameters

Parameter	Meaning
W_downhill	Post-pulse spin-conserving interlayer downhill transfer rate scale.
W_intra	Default intramaterial $c \rightarrow v$ relaxation rate scale.
W_intra_A, W_intra_B	Optional material-specific $c \rightarrow v$ relaxation rates.
t_final	Final simulation time in fs.
dt	Time step in fs.
compare_time_ref	Reference time for state-resolved difference plots. Default is pulse end.
compare_time_1, compare_time_2	Left and right comparison times for state-resolved level plots.

12.6 Plotting Parameters

Parameter	Meaning
plot_mode	Plot either changes (delta) or absolute occupations.
max_plot_points	Maximum number of plotted time points. Does not affect propagation.
delta_color_scale	Manual state-resolved color scale for Δn . If ≤ 0 , an automatic scale is used.
delta_color_norm	Either linear or symlog .
delta_color_percentile	Percentile used for automatic color scaling.
delta_colormap	Diverging colormap for state-resolved occupation changes.
level_half_length, level_x_jitter, level_linewidth, level_alpha	Appearance controls for state-resolved energy-level plots.

13 Command-Line Usage

From the repository root, run without installation by setting PYTHONPATH:

```
PYTHONPATH=src python -m abinterface --help
```

A typical simulation is

```
PYTHONPATH=src python -m abinterface \  
--N 12 \  
--pulse-duration 20 \  
--t-final 70 \  
--dt 0.02 \  
--carrier full \  
--omega-eV 2.0 \  
--lambda-soc-A 0.05 \  
--lambda-soc-B 0.05 \  
--soc-mix-cb-A 0.02 \  
--soc-mix-cb-B 0.02 \  
--tAB 0.08 \  
--W-downhill 0.01 \  
--W-intra 0.03 \  
--compare-time-1 22 \  
--compare-time-2 50 \  
--delta-color-scale 0.08 \  
--delta-color-norm linear
```

After installation,

```
pip install -e .  
abinterface-run --help
```

14 Desktop GUI Usage

The desktop GUI is based on Tkinter and can be launched by

```
PYTHONPATH=src python -m abinterface.gui
```

or, after installation,

```
abinterface-gui
```

On some Linux systems, the Tkinter bindings must be installed separately:

```
sudo apt install python3-tk
```

The GUI uses a tabbed layout:

Tab	Contents
Bands	Gaps, offsets, bandwidths, and basis size.
SOC + Interlayer	Material SOC splitting, SOC mixing, and spin-conserving interlayer hopping.
Laser	Pulse, photon energy, optical matrix elements, and optical widths.
Rates + Time	Post-pulse rates and simulation time settings.
Plotting	Color scales and state-level plot appearance.

The top bar contains the output directory, **Run simulation**, **Reset**, and **Open output**. The **Reset** button restores all parameter widgets to `ModelConfig` defaults.

15 How to Interpret Results

A physically consistent chain-mechanism run should show the following qualitative behavior:

1. During the pulse, the laser produces $A_v \rightarrow A_c$ and $B_v \rightarrow B_c$ excitations.
2. After the pulse, $A_c \uparrow$ can transfer population into $A_c \downarrow$ through material-internal SOC mixing.
3. Spin-conserving interlayer hopping moves $A_c \downarrow$ population into $B_c \downarrow$.
4. Intramaterial $c \rightarrow v$ relaxation transfers $B_c \downarrow$ population into $B_v \downarrow$.
5. Occupation buildup in $B_c \downarrow$ can influence other B_c channels when $B_c \uparrow$ and $B_c \downarrow$ are SOC-mixed.

The file `*_pathway_projected_occupations.png` is the most direct way to test this. The state-resolved level plot is useful for checking whether only a few band-edge states change or whether redistribution extends across multiple bands.

16 Model Limitations

This is a toy model. It is not a first-principles TDDFT, GW, or Boltzmann transport code. Important limitations include:

- no explicit k -resolved band structure;
- no microscopic electron-phonon or electron-electron matrix elements;
- no self-consistent charge redistribution or band renormalization;
- diagonal SOC splitting and SOC mixing are simplified model parameters;
- the magnetic moment is a projected spin-population proxy, not a full magnetic observable including orbital contributions.

The model is best used to reason about mechanisms, parameter sensitivity, and qualitative dynamical pathways.

17 Recommended Development Directions

Potential future extensions include:

1. adding k -resolved dispersions and k -dependent optical matrix elements;
2. deriving relaxation rates from explicit phonon or Coulomb spectral functions;
3. adding optional self-consistent energy shifts from accumulated charge;
4. separating coherent SOC precession from dephasing-controlled spin relaxation;
5. adding pathway-resolved current diagnostics between $A_c \downarrow$ and $B_c \downarrow$.